

EE364B Progress Report: Differentiable Inverse DC Optimal Power Flow

Author
EE364B Convex Optimization II, Stanford University

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1 Introduction

Grid operators publish hourly load and generation data, but rarely release the cost curves and capacity limits that determine who gets dispatched. I want to recover those parameters from public data alone. Knowing them enables counterfactual simulations, marginal emission factor (MEF) estimates, and calibrated planning models without access to confidential market filings.

I frame this as inverse optimization [1,3]: given dispatches that solve a DC optimal power flow (DC-OPF), find the cost and capacity parameters that best explain the data. The catch is that the forward problem is a quadratic program (QP) with kinks in the parameters; whenever a generator hits or leaves its limit, the gradient is undefined.

A standard approach [2] sidesteps this by fitting parameters to the KKT conditions of the observed dispatches. It works for cost recovery, but breaks down for capacity recovery: once $g_{\max}$ is free, the complementarity conditions are trivially satisfied at any feasible point, so the method gives no useful signal about capacities.

Instead, I backpropagate through the QP using implicit differentiation [5], which gives exact gradients wherever the active set is locally constant. The approach handles joint cost-and-capacity recovery and is available off-the-shelf via `cvxpylayers` [6].

2 Problem Formulation

Consider a network with N buses, L lines, and G generators. The forward DC-OPF I use is a QP in generator dispatches $g \in \mathbb{R}^G$ and line flows $p \in \mathbb{R}^L$:

$$\begin{aligned} & \underset{g,p}{\text{minimize}} && f^\top g + \frac{\lambda}{2} g^\top g \\ & \text{subject to} && A_g g + A_l p = d, \quad 0 \leq g \leq g_{\max}, \quad |p| \leq p_{\max}, \end{aligned} \tag{1}$$

where $f \in \mathbb{R}_+^G$ is the vector of linear marginal costs, $g_{\max}$ and $p_{\max}$ are generator and line capacity limits, and $d \in \mathbb{R}^N$ is the demand. The small quadratic term ($\lambda = 10^{-3}$) keeps the dispatch unique when two generators have the same cost.

Given T noisy observations $\{(d_t, \hat{g}_t)\}_{t=1}^T$, I want to recover the unknown parameters $\theta^* = (f^*, g_{\max}^*)$ by solving

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{T} \sum_{t=1}^T \|g^*(d_t; \theta) - \hat{g}_t\|_2^2 + R(\theta), \tag{2}$$

where $g^*(d_t; \theta)$ is the solution of (1) at demand d_t , and $R(\theta)$ is a regulariser (Tikhonov plus an optional ℓ_1 merit-order term). I train with Adam and a cosine learning-rate schedule over 300–600 steps.

Identifiability. Not all parameters can be recovered from dispatch data alone. The cost vector f is only identifiable up to positive shifts within the set of generators that are away from their bounds in at least one sample; generators that are always at a limit never influence the residual in (2). Similarly, a capacity $g_{\max,i}$ is only identifiable if generator i saturates at least once. I track both limits using a per-generator interior-fraction score and examine the correlation with recovery error in Section 4.

3 Methods

I implemented four estimators and compare them on the same experiments.

Ridge regression and **MLP** (two hidden layers, ReLU, width 128) both predict dispatch directly from demand. They serve as forecasting baselines that completely ignore the OPF structure and cannot recover f or $g_{\max}$.

The **KKT-residual estimator** [2, 4] fits θ by minimising the sum of squared KKT residuals (stationarity and complementarity) evaluated at the observed dispatches. This is a standard inverse-optimisation approach and is my main point of comparison.

The **differentiable estimator** (this work) solves (2) directly by backpropagating through (1) via `cvxpylayers`. I run three variants: *caps fix* (capacities held at their true values, to isolate cost recovery); *full* (joint cost and capacity recovery); and *strat.* (full recovery with a Laplacian-smoothed $24 \times G$ cost table to capture diurnal merit-order shifts).

For evaluation I use five metrics: clean-dispatch NRMSE on held-out demands; cost cosine $\cos \hat{f}$ and Spearman rank correlation; pairwise merit-order accuracy (fraction of generator pairs whose cost ordering is correct); and capacity cosine $\cos \hat{g}_{\max}$. Cosine and merit-order accuracy matter most here because a positive scaling or shift of f leaves the dispatch solution unchanged, so there is a fundamental indeterminacy that RMSE on f would not reflect.

4 Preliminary Results

4.1 Main Experiments

A random 10-bus network ($G = 10$, $L = 12$, $T = 200$, relative noise 0.05, three seeds) gives Table 1. The differentiable estimator with fixed capacities recovers cost cosine 0.959 versus 0.908 for KKT (+0.051, 37% lower seed std). Joint capacity recovery costs one cosine point ($0.959 \rightarrow 0.949$) but recovers $\cos \hat{g}_{\max} \geq 0.999$ versus 0.870 for KKT; this is the clearest evidence that differentiating through the QP is worth the added complexity.

Table 1: Headline Comparison, 10-Bus Synthetic Network. $T = 200$, three seeds, mean $\pm$ std. Ridge and MLP are forecasting baselines and do not estimate f .

Method	NRMSE _{cl.}	$\cos \hat{f}$	Merit acc.	$\cos \hat{g}_{\max}$
Ridge	0.110	—	—	—
MLP	0.477	—	—	—
KKT	0.138	.908 $\pm$.019	.881 $\pm$.034	.870 $\pm$.055
Diff., caps fix.	0.125	.959$\pm$.012	.904$\pm$.013	1.00
Diff., full	0.208	.949 $\pm$.018	.889 $\pm$.022	.999
Diff., strat.	0.198	.947 $\pm$.018	.874 $\pm$.046	.999

For MEFs I computed $\partial(c^\top g^*)/\partial d$ at 10 operating points via autograd and compared against ground-truth (Table 2). Cosine 0.987 is the strongest result in the project; the gradient structure is preserved even where $\hat{f}$ is only recovered up to scale.

Table 2: MEF Accuracy, 10 Operating Points.

Estimator	MEF cosine	MEF RMSE
Diff., full	0.986	0.070
Diff., caps fix.	0.987	0.064

For diurnal data (24-hour merit-order rotation, renewables cheap by day, peakers cheap at dusk) the static estimator recovers $\cos \hat{f} = 0.963$ (NRMSE 0.365) while the stratified estimator gets $\cos \hat{f} = 0.956$ but also recovers the full hour-resolved cost table at $\cos \hat{F}_{\text{table}} = 0.935$, which the static model cannot represent.

On an 8-fuel single-bus PJM-like model (198 GW; load 80–130 GW; $T = 400$; one seed) the estimator gives $\cos \hat{f} = 0.672$, Spearman 0.810, merit-order accuracy 0.786 (22 of 28 fuel-pair orderings correct). The result rests on a single seed; I will add two more before drawing conclusions.

5 Next Steps

The main comparison (Section 4.1) and MEF result are in good shape for the final report. The remaining work is about a few open issues I need to resolve before submitting.

For the noise sweeps, my best guess is that the noise gets clipped or rescaled somewhere in the data pipeline before it reaches training, so the estimator never actually sees different noise levels. Once I find and fix that I will rerun and check whether the cost cosine drops meaningfully with σ . For the identifiability experiment, I want to replace the interior-fraction proxy with a score based on the rank of the active-set indicator matrix, which is more directly connected to the identifiability argument. I also want to figure out the finite-difference discrepancy; I plan to sweep the step size ε and rescale the parameters so the Jacobian entries are comparable in magnitude.

Beyond those fixes, I would like two more seeds for PJM to get error bars, a test on a 39-bus IEEE network, and a sweep of the regulariser and Laplacian smoothing weights for the stratified estimator.

References

- [1] R. K. Ahuja and J. B. Orlin, “Inverse Optimization,” *Operations Research*, vol. 49, no. 5, pp. 771–783, 2001.
- [2] A. Keshavarz, Y. Wang, and S. Boyd, “Imputing a Convex Objective Function,” in *IEEE International Symposium on Intelligent Control*, 2011, pp. 613–619.
- [3] P. M. Esfahani, S. Shafieezadeh-Abadeh, G. A. Hanasusanto, and D. Kuhn, “Data-Driven Inverse Optimization with Imperfect Information,” *Mathematical Programming*, vol. 167, no. 1, pp. 191–234, 2018.
- [4] A. Aswani, Z.-J. Shen, and A. Siddiq, “Inverse Optimization with Noisy Data,” *Operations Research*, vol. 66, no. 3, pp. 870–892, 2018.
- [5] B. Amos and J. Z. Kolter, “OptNet: Differentiable Optimization as a Layer in Neural Networks,” in *International Conference on Machine Learning*, 2017, pp. 136–145.
- [6] A. Agrawal, B. Amos, S. Barratt, S. Boyd, S. Diamond, and J. Z. Kolter, “Differentiable Convex Optimization Layers,” in *Advances in Neural Information Processing Systems*, 2019, pp. 9558–9570.
- [7] S. Diamond and S. Boyd, “CVXPY: A Python-Embedded Modeling Language for Convex Optimization,” *Journal of Machine Learning Research*, vol. 17, no. 83, pp. 1–5, 2016.
- [8] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University Press, 2004.